

The image features a central dark mauve rectangle containing the text. This rectangle is framed by delicate floral illustrations in the corners. The top-left and bottom-right corners have small clusters of pink flowers with green leaves. The left and right sides are adorned with larger, more detailed pink flowers and buds, some with green leaves, creating a symmetrical floral border.

AI FIT TRACK

TN SKILLS



Team members



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Project Overview

What We Developed

- * AI FitTrack is a web-based fitness application.
 - * It combines fitness tracking with Artificial Intelligence.
 - * The application collects fitness-related information from users.
 - * AI analyzes the information and generates recommendations.
 - * We developed the project using a
frontend + backend + database + AI API architecture.
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Technology Stack:

Frontend → Node.js Backend → MongoDB

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Gemini API**



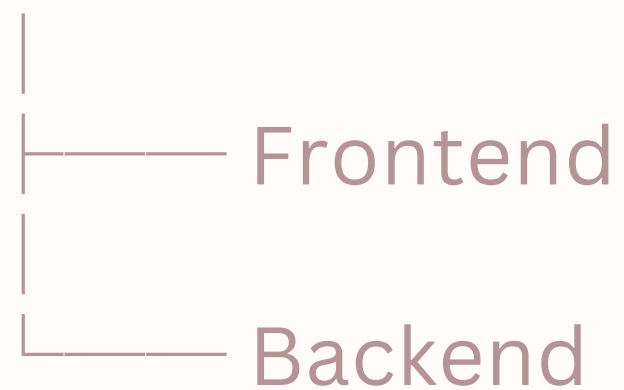


Project setup in vs code

Step 1 – Creating the Project

- * We created the project folder.
- * We opened the project in Visual Studio Code.
- * We organized the project into frontend and backend sections.
- * We created the required files and folders.
- * We used the VS Code terminal to install packages and run the application.

AI-FitTrack



Npm setup

Step 2 – Initializing NPM

NPM (Node Package Manager) is used to manage the packages required for our Node.js project.

Process:

1. Open the project in VS Code.
2. Open the terminal.
3. Navigate to the backend folder.
4. Initialize the Node project.
5. Install the required dependencies.

```
npm init -y
```

```
npm install express
```

```
npm install mongoose
```

1. package.json is created.
2. It stores project information and dependencies.



Node. Js Backend

Step 3 – Creating the Server

- * We used Node.js to develop the backend.
- * We used Express.js to create the server.
- * The server receives requests from the frontend.
- * It processes the data and sends responses.
- * We configured a port for running the backend.

Backend Flow:

Frontend Request → Express Server → Processing → Response

Important :

Server running on port 5000



MongoDB Integration



Step 4 – Connecting MongoDB

- * We selected MongoDB as the database.
- * MongoDB stores application data in collections and documents.
- * We used Mongoose to connect Node.js with MongoDB.
- * The backend communicates with MongoDB to store and retrieve data.

Connection Flow:

Frontend → Node.js → Mongoose → MongoDB

We also tested the database connection and resolved connection errors during development.



Creating API/Backend Routes

Step 5 – Creating Backend APIs

We created backend routes to allow the frontend to communicate with the server.

The backend handles:

- * User information
- * Fitness data
- * Fitness goals
- * Recommendation requests
- * Database operations
- * User enters data



Frontend sends API request



Node.js receives request



Backend processes data



Re



gemini API Integration

Step 6 – Adding Artificial Intelligence

- * We integrated the Gemini API into AI FitTrack.
- * We generated an API key for accessing the AI service.
- * The user's fitness information is provided to the AI.
 - * Gemini processes the information.
- * The AI-generated recommendation is returned to our application.

AI Flow:

User Data → Backend → Gemini API → AI Response → Frontend
This is the main feature that makes our application AI-based.

Creating the Recommendation Feature

Step 7 – AI Recommendation

We created a recommendation section in the application.

The user provides information such as:

- * Fitness goal
- * Age
- * Weight
- * Height
- * Activity level
- * Other required details

The application sends the information to the AI.

Output:

The AI generates a fitness-related recommendation that is displayed in the recommendation box.



Testing & Debugging

Step 8 – Testing the Complete System

We tested each part individually and then tested the complete application.

We checked:

- * Frontend interface
- * NPM packages
- * Node.js server
- * API routes
- * MongoDB connection
- * Gemini API
- * User input
- * AI recommendation output

Debugging Process:

Error → Find the Cause → Modify Code → Run Again → Test



final project workflow

Complete Working Process

USER



Enter Fitness Data



FRONTEND



NODE.JS / EXPRESS



MONGODB GEMINI API



Store Data AI Analysis



RECOMMENDATION



USER OUTPUT

Our project combines:

Frontend + Node.js + Express + MongoDB + Gemini API

Final output & Conclusion

Final Output

Our completed application provide

- * Fitness information inp
- * Fitness goal selectio
- * AI-powered recommendatio
- * Backend processin
- * Database connectivit
- * User-friendly interfac

What We Learn

- * Project setup using VS Co
- * NPM and package installatio
- * Node.js and Expres
- * MongoDB database connectio
- * API integratio
- * Gemini AI integratio
- * Testing and debuggin
- * Full-stack application developmen

Thank You

Team AI FitTrack

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Thank you



